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Title: Sensitivity Analysis on a Novel Continuous Particle Swarm Optimization Algorithm for EMA Model Updating

Author(s): Giuseppe Mugnos^{*1}, Francesco Lo Iacono¹, Giacomo Navarra¹, Angela Ricciardello¹.

¹ Faculty of Engineering and Architecture, University of Enna Kore, Enna, Italy,

Emails: giuseppe.mugnos@unikorestudent.it, francesco.loiacono@unikore.it, giacomo.navarra@unikore.it, angela.ricciardello@unikore.it.

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Abstract

The effectiveness of design, construction, and validation processes in the production of human artifacts are crucial to ensure success in the development of an industrial product and to guarantee its structural robustness during lifetime. More often, these three processes are not related one to each other and the discrepancy between the design of an artifact and its actual construction can generate uncertainty on the mechanical behaviour of the structures. To fill up the gap between design and actual built structural models, several model updating methods [1] have been proposed by the scientific literature in the past decades.

In a previous work proposed by the same authors [2], a structural model updating procedure based on a recent variant of the Particle Swarm Optimization algorithm, called Continuous Particle Swarm Optimization (CPSO), has been developed showing good performance in terms of accuracy and computational time reduction in comparison with other heuristic search algorithms.

In the present paper, the results of a numerical campaign aiming to assess the robustness of the proposed algorithms CPSO is reported. An extensive sensitivity analysis has been conducted by varying the CPSO parameters and the target objective functions by focusing on several structural parameters.

The proposed model updating method has been stressed and validated by using the experimental data recorded on a real-scale large electrical device whose dynamical properties have been characterized by using Experimental Modal Analysis (EMA) approach. Both random noise and seismic experimental responses of the structure under investigation have been acquired by taking advantage of the shaking table system of the L.E.D.A. (Laboratory of Earthquake engineering and Dynamic Analysis) Research Institute at the “Kore” University of Enna, Italy [3].

References

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